Google Gemini Integration

Vertex AI Studio and Working with the Gemini API

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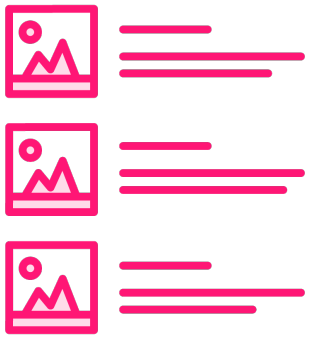
Working with Vertex AI Studio and Google AI Studio

Gemini 2.0 Innovations

Native tool use



Multimodal output

Real-time streaming

Google AI Studio vs. Vertex AI Studio

Google AI Studio	vs.	Vertex AI Studio
Target developers, researchers, beginners exploring Gemini models Focus on prototyping and experimentation Limited LLMOps or MLOps capabilities		Target developers and engineers working in an enterprise environment Focus on prototyping but inside the context of the enterprise Greater set of capabilities from the wider Vertex AI platform

Demo: Customize Gemini with Prompting and Finetuning on Vertex AI

Integrating Gemini into Applications

Google Gen AI SDK

```python
from google import genai

API_KEY = "YOUR_API_KEY"

client = genai.Client(vertexai=False, api_key=API_KEY)

response = client.models.generate_content(
    model="gemini-2.0-flash-001",
    contents="""Explain bubble sort to me.""",
)

print(response.text)
```

Google Gen AI SDK

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```

Use Application Default Credentials

```
> gcloud auth application-default login
```

Demo: Using a Gemini Agent with Google Search and Code Execution

Working with Frameworks and Tools

LangChain / LangGraph

Llama Index

Crew AI

LiteLLM

OpenRouter